

Mobile Apps

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Reality-based Interaction

Mobile interaction breaks the assumption that interaction means a user is primarily stationary.

Multimodal interaction breaks the assumption that interaction with computer systems primarily involves providing input via one modality and receiving output via another modality, such as a visual display.

Reality-based Interaction

Ubiquitous computing breaks the assumption that users must command computers to get things done. Ubiquitous computing exploits context sensing to enhance and automate interactions for the user.

Mixed Reality and tangible interaction break the assumption that virtual reality (VR) and physical reality are separate.

Mobile user interfaces



Mobile user interfaces

Mobile implies smaller sizes (devices, screens, etc.)

Design for mobile interfaces emphasizes visual hierarchy.

A visual hierarchy ranks graphical elements and places them in groups so that high-level elements contain low-level elements.

A well-designed hierarchy helps the user understand the order in which a task can be completed (i.e., the task flow)

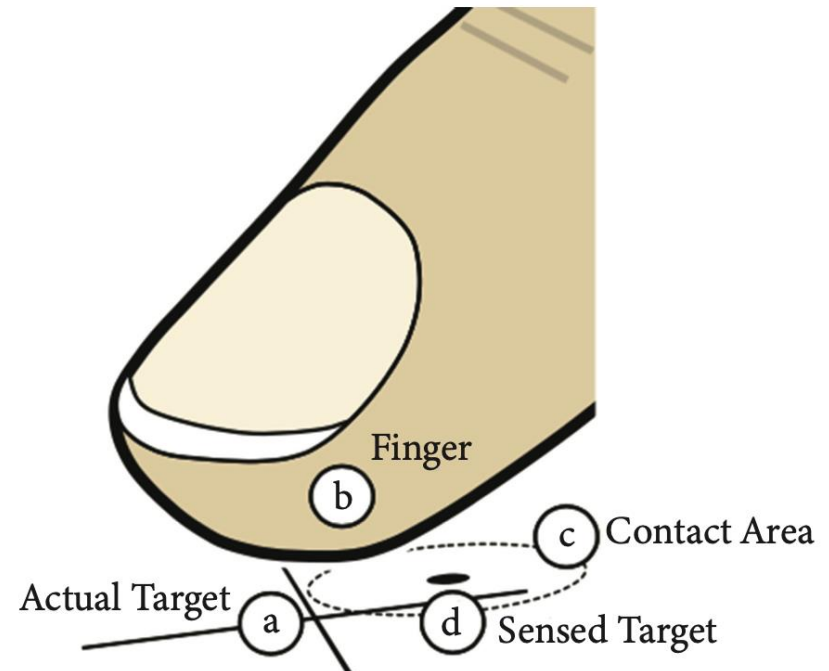
See more one: <https://www.figma.com/resource-library/what-is-visual-hierarchy/>
<https://www.interaction-design.org/literature/article/visual-hierarchy-organizing-content-to-follow-natural-eye-movement-patterns>

Mobile user interfaces

Small-screen interaction

Fingers occlude the display.

Interaction techniques can help!



Mobile user interfaces

A wide variety of sensors allows inferences to be made about the user's context and to take information into account in interaction.

Examples:

- location sensing (e.g., GPS, wireless radio-based methods),
- gesture and motion sensing (e.g., accelerometer and gyroscope),
- audio and speech sensing (microphone),
- light sensing (e.g., to detect if the phone is held against the ear).

Supporting mobile interaction means supporting interaction that takes place while in physical transition between places.

Mobile context refers to relevant factors in a situation that may affect interaction with the computer.

Multimodal Interfaces

“Multimodal systems process two or more combined user input modes— such as speech, pen, touch, manual gestures, gaze, and head and body movements— in a coordinated manner with multimedia system output.”

Oviatt, S., 2012. Multimodal interfaces. In *The human-computer interaction handbook* (pp. 439-458). CRC press.



<https://youtu.be/6uvFrDddYnM>

Ubiquitous computing

M.Weiser.The computer of the 21st century. Scientific American, 265(3):94–104, 1991.

"Ubiquitous computing names the third wave in computing, just now beginning. First were mainframes, each shared by lots of people. Now we are in the personal computing era, person and machine staring uneasily at each other across the desktop. Next comes ubiquitous computing, or the age of calm technology, when technology recedes into the background of our lives."

Ubiquitous Computing

“The most profound technologies are those that disappear. (...) Consider writing (...) is ubiquitous in industrialized countries Not only do books, magazines and newspapers convey written information, but so do street signs, billboards, shop signs and even graffiti.”

“...a new way of thinking about computers in the world, one that takes into account the natural human environment and allows the computers themselves to vanish into the background.”

Ubiquitous computing

Embedded input and output: sensors (input) and output devices embedded in objects or environment.

Context Awareness: a computer system is context-aware “if it uses context to provide relevant information and/or services to the user, where relevancy depends on the user’s task”

- social environment of the user
- physical environment
- activity recognition

interactions should be natural in its context and implicit (vs GUIs, users give explicit commands)

Context-Aware Computing

Context: situational elements relevant to interaction between user, application, environment

Context-awareness: situationally appropriate; apps adapting to context, increasing value to users

“where you are,
who you are with,
and what resources are nearby”

B. Schilit, N. Adams, and R. Want. 1994. Context-Aware Computing Applications. In *Proceedings of the 1994 First Workshop on Mobile Computing Systems and Applications (VMCSA '94)*. IEEE Computer Society, Washington, DC, USA, 85-90.
DOI=<http://dx.doi.org/10.1109/VMCSA.1994.16>

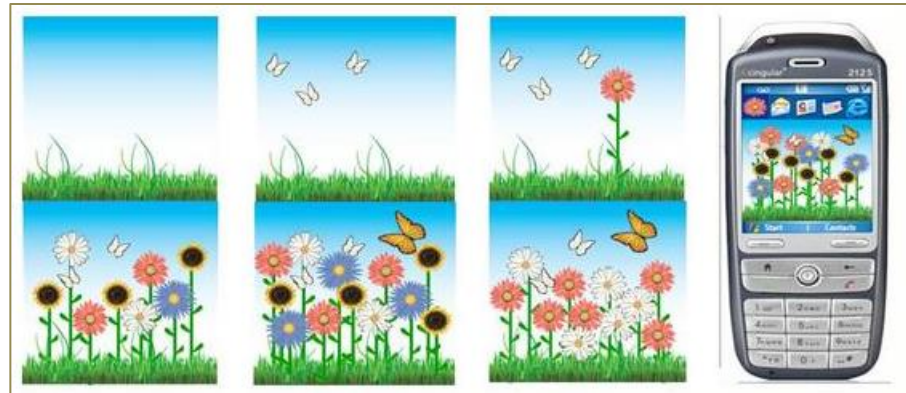
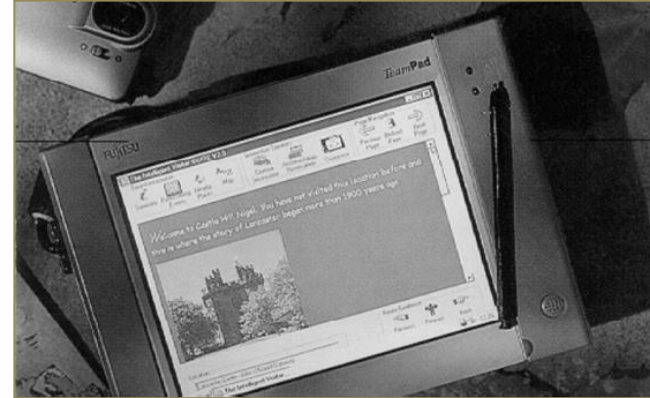
Context-Aware Computing

Context-awareness: situationally appropriate; apps adapting to context, increasing value to users

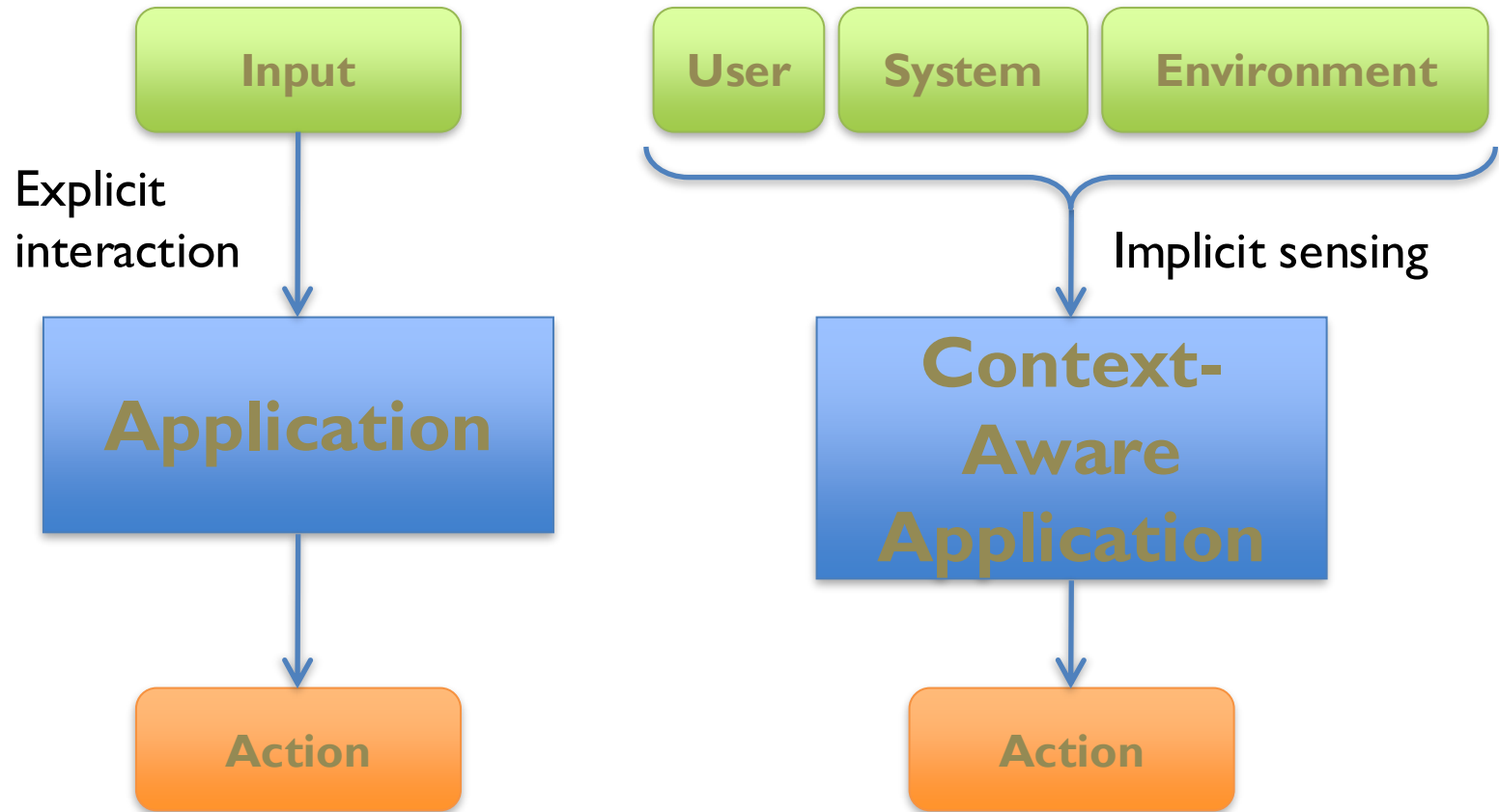
Using sensors and actuators to improve human-computer interaction and (computer-mediated) human-human and human-environment interaction

Context-Aware Computing: Examples

Home monitoring, Tour guide, Reminders, Persuasive displays



Context-Aware Computing



20+ years of context-aware computing research

Driven by a single problem

Reminder to buy milk

When to deliver: not time/location specific

How to deliver: appropriate modality



Context-Aware Applications

What is a context-aware application?

App that uses context to perform some behavior/service for its user(s)

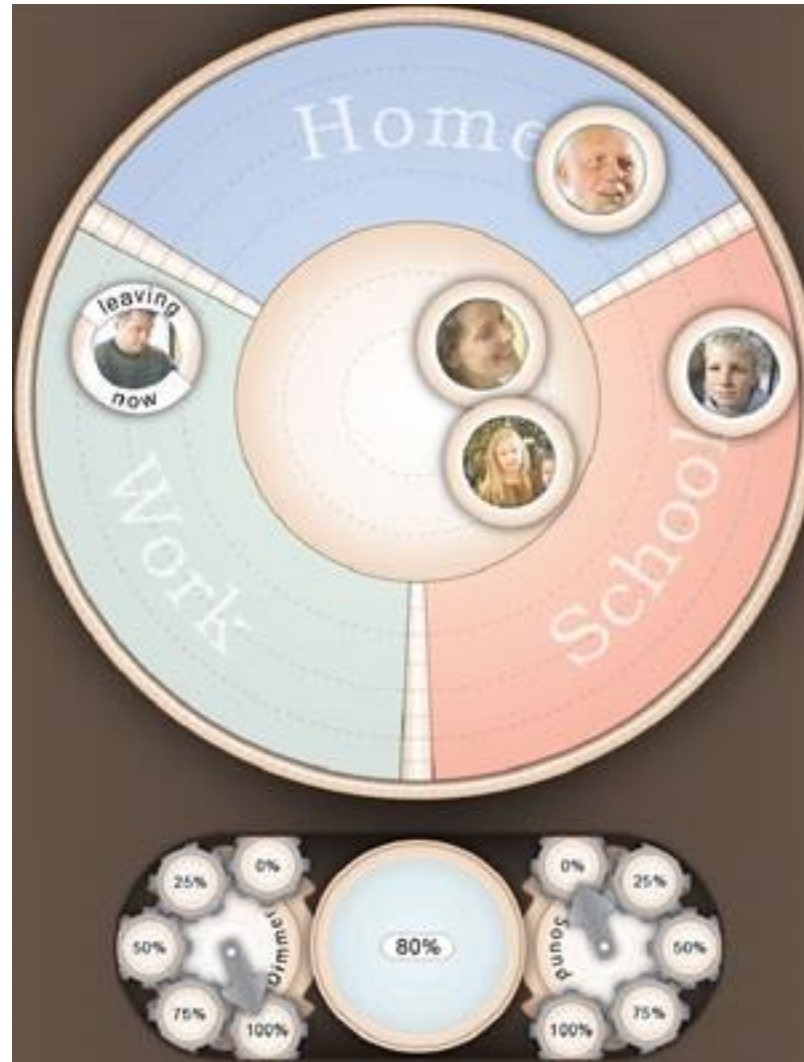
3 types of behaviors:

Display context

Automatically execute/adapt services

Tag captured information for easier retrieval

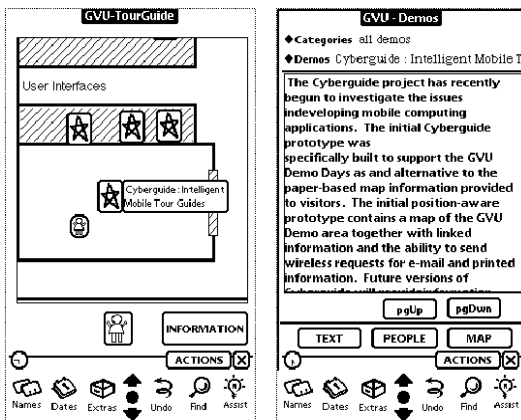
Directly displaying context



Auto Execute/Adapt Services

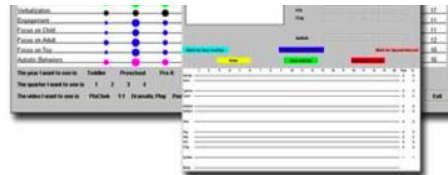
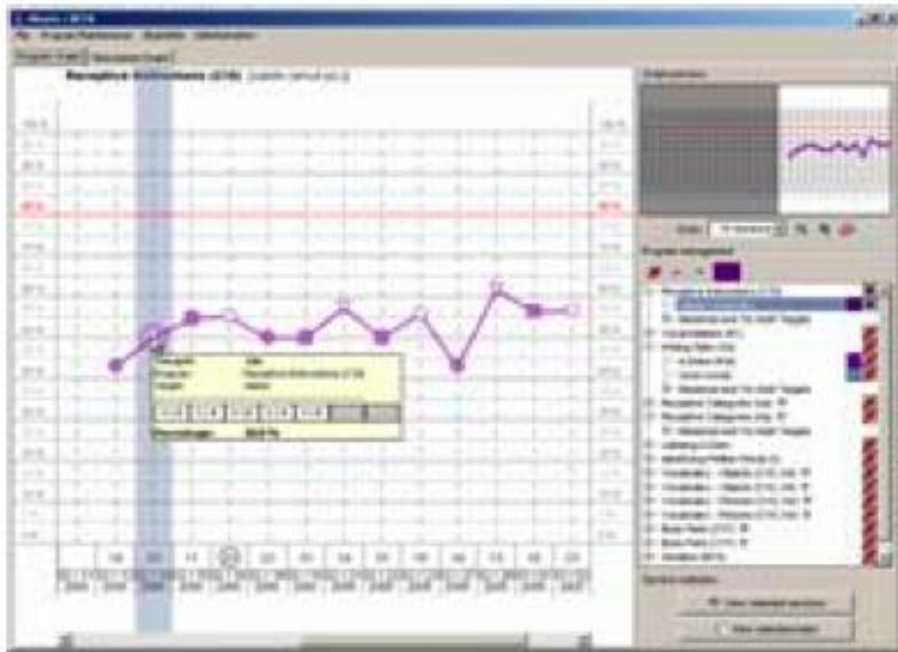
Select and perform a relevant service from multiple services

Change how a service behaves or is executed



Tag Information

Use context to aid in context-based retrieval



What are the Impediments?

Cost and difficulty in deploying infrastructure

Applications are of limited value

- Not modeling the right tasks

Applications lacking in usability

- Apps lacking in support for end-user control
- Apps lacking in intelligibility

Focus on High-Level Activity

Not where you are, but what you're doing

- Physical activity: personal informatics, quantified self
- Cognitive abilities
- Changing capabilities over time

Mixed Reality

User interface technology that mixes virtual and real content to create a new, augmented experience that is interactive and appears authentic to the user:

- (1) a programmatic association between virtual content and the real world
- (2) an interactive display of that association.

Mixed Reality



Milgram and Kishino's (1994) Reality-Virtuality Continuum

Mixed Reality

Virtual Reality (VR): interactive environment that responds to the user's actions, for example, via locomotion or camera control.

Augmented Reality (AR): adds real-world content to a virtual environment (augmented virtuality) or virtual content to a real-world experience.



Augmented Reality - Mobile

<https://youtu.be/Xx9Z57jhrYw>

Challenges:

Natural Light Reflection

Arm Fatigue

Holding Device + Touch Interactions